Sepsis Risk at ED Triage — Abstracts

Oct 2, 2026 · @Rupesh Ghimire

Title options and one-line pitch

Recommended title: **Spotting sepsis in the waiting room: explainable risk prediction from triage data alone.**

Style	Title
Plain and direct (recommended)	Spotting sepsis in the waiting room: explainable risk prediction from triage data alone
Formal	Explainable machine learning for early sepsis risk at emergency department triage: development and internal validation against rule-based screens
Question-led	Who is quietly getting sicker? Interpretable sepsis risk estimation at ED triage

One-line pitch: We test how well the first few minutes of triage information can flag sepsis risk, and we show the reasons behind every flag.

Use "motivated by SAFE-WAIT" in any title or subtitle, never "SAFE-WAIT model" or "replicating SAFE-WAIT". The project does not use that model or its data.

Humanised abstract

For judges, clinicians and the public. 275 words, no jargon, no results claimed.

Sepsis can turn a treatable infection into organ failure within hours, and it often begins quietly. In a crowded emergency department, someone who looked stable at triage can be getting sicker in the waiting room while staff attend to more obviously urgent cases. NSW Health's SAFE-WAIT program was built for exactly this gap, and from 2027 it will be trialled across five Western Sydney hospitals.

Our project asks a simple question: how much can the first few minutes of information tell us? We use only what a triage nurse already records, such as temperature, heart rate, breathing rate, blood pressure, age, the reason for the visit and how the patient arrived, to

estimate each person's risk of sepsis. No blood tests and no new devices are needed.

We will train and compare three models, from a simple and transparent one to a more flexible one, on about 425,000 de-identified emergency visits from a public research database. We will check them against the screening rules used today, and measure workload as well as accuracy: how many patients would be flagged, and how many of those truly have sepsis.

A score that nobody understands will not be trusted, so every prediction comes with its reasons. We will also test whether the model works equally well for older and younger patients, for women and men, and for other groups.

The result is a research prototype that shows a low, moderate or high risk, like a traffic light. It is not ready for clinical use. The next step is to test it on de-identified NSW hospital data, with ethics approval, to see whether it holds up for local patients.

Technical structured abstract

For the conference submission. 325 words, proposal form, written to be true before any results exist.

Background. Emergency department (ED) crowding delays recognition of sepsis in patients awaiting assessment. NSW Health's SAFE-WAIT program applies a gradient-boosted model at triage; its retrospective evaluation of 108,401 Western Sydney presentations (preprint) reported earlier antibiotic administration for patients flagged moderate or high risk. Open, reproducible evidence on what triage-only data can achieve, with explanations and subgroup performance, remains limited.

Aim. To develop and internally validate explainable machine learning models that estimate sepsis risk from information available at ED triage, and to compare them with rule-based screens.

Methods. Retrospective cohort study using MIMIC-IV-ED v2.2 (about 425,000 ED visits, 2011 to 2019) linked to MIMIC-IV. Adults with at least one triage vital sign are included. The primary outcome is sepsis coded in the ED or the linked admission; the secondary outcome is in-hospital death or ICU transfer within 12 hours. Predictors are restricted to triage: vital signs, age, sex, chief complaint, arrival mode, arrival time and prior-encounter history. Logistic regression, random forest and XGBoost are tuned by patient-grouped five-fold cross-validation on a 70% training split, calibrated on a 15% validation split and evaluated once on a 15% test split. We report AUROC, AUPRC, calibration, and sensitivity, positive predictive value and alert rate at pre-specified cut-offs, with 95% bootstrap

confidence intervals. Models are compared at matched alert rates with triage versions of SIRS, qSOFA, NEWS and nurse-assigned acuity. Decision curve analysis assesses net benefit, SHAP values explain predictions, and performance is audited by sex, age, recorded race, arrival mode and data completeness. Reporting follows TRIPOD+AI.

Expected outputs. A calibrated three-tier risk prototype; a quantified comparison of models and rules; the added value of triage context beyond vital signs; and a subgroup performance audit.

Significance. The study offers a transparent benchmark aligned with a current NSW Health priority. Findings come from one US centre and are not evidence of performance in NSW. External validation on de-identified NSW ED data, under ethics and data custodian approval, is the planned next stage.

Short version, 142 words, for forms with a tight limit

Crowded emergency departments can delay recognition of sepsis in patients waiting for assessment. Motivated by the clinical aim of NSW Health's SAFE-WAIT program, this project develops explainable machine learning models that estimate sepsis risk from information available at triage: vital signs, age, sex, presenting complaint, arrival mode and prior history. Using about 425,000 de-identified ED visits in MIMIC-IV-ED, we will compare logistic regression, random forest and XGBoost with triage versions of SIRS, qSOFA and NEWS on a patient-level held-out test set. Performance will be reported as AUROC, AUPRC, calibration, and sensitivity and positive predictive value at matched alert rates, with bootstrap confidence intervals. SHAP values will explain individual predictions, and performance will be audited across patient subgroups. The output is a three-tier research prototype, not a clinical tool. Validation on de-identified NSW data, subject to ethics approval, is the planned next step.

Results-ready version

For the poster and any final submission. Every bracket is filled from `results_sentences.md`, which the pipeline writes after the test set is opened. Do not fill a bracket from any other run.

Background. Crowding delays recognition of sepsis in patients waiting for ED assessment. We tested how well information available at triage alone can identify sepsis risk, with explanations for each prediction.

Methods. Retrospective cohort of adult ED visits in MIMIC-IV-ED v2.2. The outcome was sepsis coded in the ED or linked admission. Logistic regression, random forest and XGBoost were trained on triage vital signs, demographics, chief complaint, arrival mode and prior history, and evaluated once on a patient-level held-out test set against triage versions of SIRS, qSOFA and NEWS.

Results. Of [N] visits by [n] patients, [n] ([x]%) had sepsis recorded. In the test set ([n] visits, [n] cases), XGBoost achieved AUROC [x] (95% CI [a] to [b]) and AUPRC [x] ([a] to [b]); logistic regression achieved AUROC [x] ([a] to [b]). At the same alert rate as triage SIRS ([x]% of arrivals), the model identified [x]% of sepsis cases against [y]% (difference [d] points, 95% CI [a] to [b]). Adding complaint, arrival and history to vital signs changed AUROC from [x] to [y]. After calibration the slope was [x]. The high-risk tier ([x]% of arrivals) held [x]% of sepsis cases at [n] reviews per case found. The strongest contributors were [feature 1], [feature 2] and [feature 3]. The share of cases flagged ranged from [x]% to [y]% across patient subgroups.

Conclusion. Triage data alone [did / did not] identify sepsis risk better than rule-based screens at an equal alert burden, and the main drivers were clinically recognisable. These results are from one US hospital with a code-based outcome. Validation on de-identified NSW data is required before any local claim.

Three ways the conclusion may need to change

- If the gain over logistic regression is small, say that a simple, readable model is nearly as good, and that this favours transparency.
- If the model does not beat nurse-assigned acuity, say so, and frame the model as a second check, not a replacement.
- If a subgroup shows a real gap, name the group and the size of the gap in the conclusion.

Keywords, word counts and submission notes

Keywords: sepsis; emergency department; triage; machine learning; explainable artificial intelligence; clinical decision support; risk stratification; XGBoost; SHAP; algorithmic fairness.

Version	Words	Use it for
Humanised	275	Lay summary, poster handout, speaking notes

Version	Words	Use it for
Technical structured	325	Conference submission with a 350-word limit
Short	142	Forms with a 150-word limit
Results-ready	About 250 once filled	Poster and final proceedings

What changed from your draft, and why

- "Publicly available emergency department datasets" is now named: MIMIC-IV-ED v2.2. Reviewers look for the dataset.
- The outcome is defined. "Sepsis risk" alone leaves the reader guessing what was predicted.
- Rule-based comparators are added. A model compared only with other models does not show clinical value.
- Alert rate and positive predictive value are added beside sensitivity and AUROC. These are the numbers a nurse manager asks about.
- The limits are stated in the abstract itself: one US centre, not evidence for NSW.

Before you submit

- [ ] Check the word limit and required headings in the competition's call for abstracts
- [ ] Trim to fit: the first sentence of Background and the Expected outputs paragraph can go first
- [ ] Confirm with your supervisor that the SAFE-WAIT wording is acceptable
- [ ] Add authors, affiliations and the supervisor's name
- [ ] Check the competition's rule on declaring AI assistance in writing, and declare it if required
- [ ] Keep the proposal version for submission; swap in the results-ready version only after the test set is opened

Sources for the facts used above: SAFE-WAIT evaluation preprint; NSW Government release, 13 September 2026; MIMIC-IV-ED v2.2.